Chapter 9

Examples of Professional Trading Strategies

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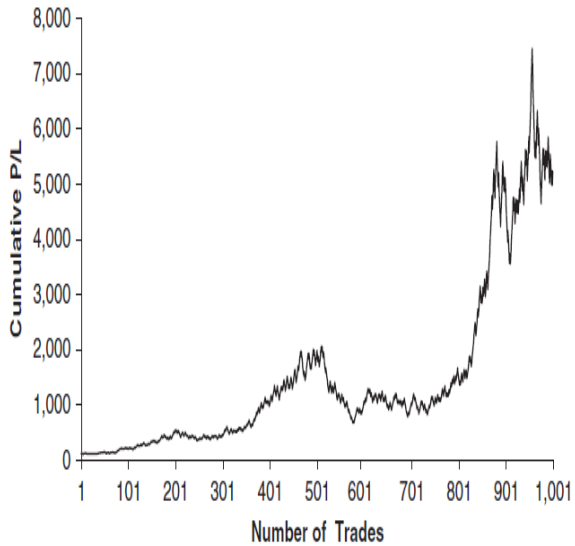
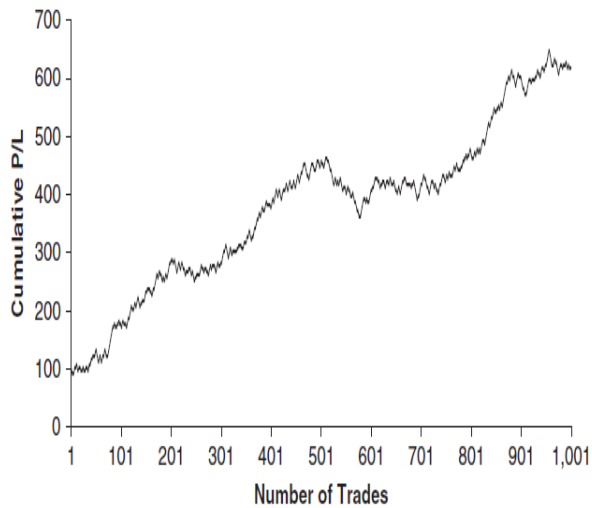
9.1. Kelly criterion in money management

9.1.1. Introduction

The presentation in this section is taken from Sinclair (2013), chapter 8. We begin with an example. Let's say you have $100, and you play a game where you flip a coin with probability of 0.5 for heads and 0.5 for tails. For heads, you win 2 dollars for each dollar bet, i.e. you end up with a total of $3 after the bet. For tails, you lose your dollar bet. How much do you want to bet each time in order to optimize your earnings? Typically people use one of three possible methods:

- Trade according to "feel", i.e. change the size each time depending on the trader's view.
- Trade a fixed amount every time, e.g. $5.
- Trade a fixed fraction of the total portfolio every time.





The results can be very different. The above diagrams show some simulations about possible bankrolls with different trading strategies (from Sinclair (2013), p.132). The left diagram represents the strategy with a constant bet of 5% of the bankroll, with a final wealth of $5207. The right diagram is a bet of $5 for each trade, with a final wealth of $620. The winning percentages are 550 bets out of 1000 in each case.

Intuitively, if you bet all your money on each game, you will easily lose a game and be bankrupt. However, if you bet too little, you will not make too much money. A controversial method known as the **Kelly criterion** has been derived and named after John Kelly, a physicist in Bell Labs, in 1956. Originally the method was developed in the context of information theory. Subsequently it has been popularized for blackjack gambling in casinos, as well as being used for trading in hedge funds.

9.1.2. Derivation

We assume that the initial wealth level = W_0. In a game, when we win we will earn a%, and when we lose, the loss is b%. Furthermore, assume that each bet is a fraction f of the total current wealth. In this set up, after n wins and m losses, we will have:

Total wealth $W = W_0 \text{ x } G(f)$, where gain factor $G(f)$ is defined as

$$G(f) = (1+fa)^n(1-fb)^m$$

or on a per trade basis

$$G(f)^{[1/(n+m)]} = g(f) = (1+fa)^p(1-fb)^q$$
$$p = n/(n+m);\ q = m/(n+m)$$

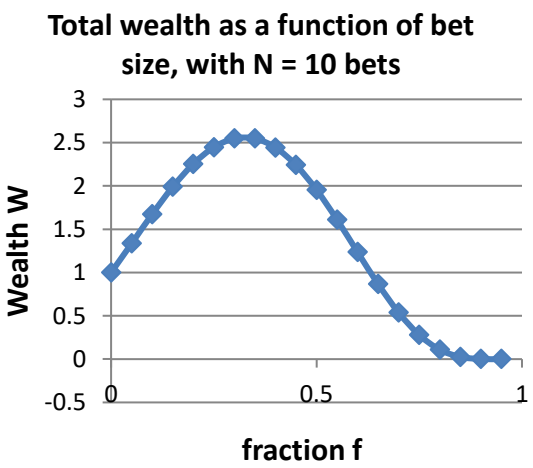
The next question is: what should be the objective function in the optimization process? It is obvious that not all objective functions are desirable. For example, we can examine the strategy for *maximizing the expected return*. Using the numbers from the earlier game, suppose that your bankroll is 1 dollar and in order to maximize the expected (mean) return, the correct strategy is for you to bet everything (f = 1). Let's say that the game goes on for 10 rounds. After 10 rounds, there is one chance in 1024 (=2^{10}) that you will have 59,049 dollars (=3^{10}, assuming that you win every time and triple the bankroll) and 1023 chances out of 1024 bets that you will have 0 dollars. Thus your expected wealth is 59049/1024 = 57.67 dollars (which is the arithmetic mean). We can see that the median wealth is 0 dollar. While the expected wealth is the highest, the risk is also very high as there is a very big possibility that you would end up with nothing.

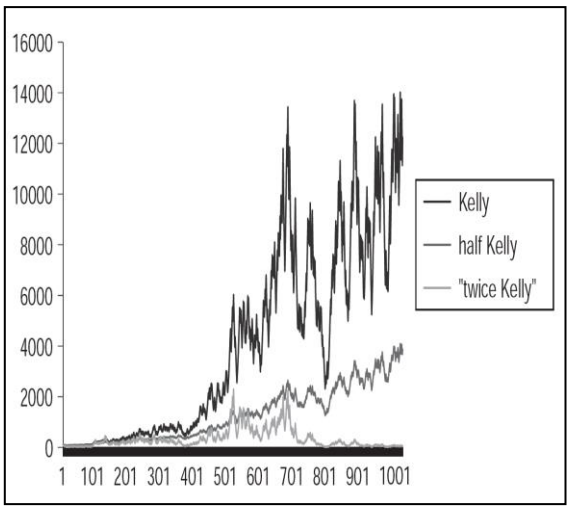
Instead of maximizing expected wealth, one can maximize a utility function, say a log utility (c.f. section 4.3.5.3). We can write: $\mathrm{Ln}[g(f)] = p\ln(1+fa)+q\ln(1-fb)$. The maximum of this function can be found by differentiating w.r.t. f and set the derivative to 0:

$$pa/(1+fa) - qb/(1-fb) = 0$$
$$f = (pa - qb)/ab$$

This fraction f is the Kelly criterion. The expected wealth after N bets is:

$$W = W_0[1+ p\ln(1+fa)+q\ln(1-fb)]^N$$





The diagram on the left shows the result when p = 0.55, q = 0.45, a = 2, b = 1, N = 10; Kelly ratio f is 0.325. In general, if we bet more than the Kelly ratio, the result is worse. In this example, the wealth will be lower than the initial value when f > 0.632. A simulation example is taken from Sinclair (2013), p.137. We can see that betting with the Kelly ratio gives the best result. Otherwise the wealth will be higher if each bet is based on "half Kelly" rather than "twice Kelly," and the difference can be quite substantial.

The case for stocks

In the stock market setting, the outcome is not binary, but will consist of a continuous range of values. The profit / loss of a trade depends on how much the stock price has changed compared to the starting level. Therefore, the profit/loss profile would also follow some kind of distribution.

In order to formulate this problem, we can start with modeling the stock price by a certain probability distribution, e.g. the Geometric Brownian motion as used in the derivation of option formulas. Using the previous notation, the gain factor can now be written as:

$$G(f) = \prod_{i=1}^{N} [1 + fh(X_i)]^{p_i}$$

where X_i is a random variable, p_i is the probability of having a payoff, and $h(X_i)$ is the payoff. The expected value of the log of the gain factor is:

$$E[ln(G(f))] = \sum_{i=1}^{N} [p_i \, ln(1 + fh(X_i))]$$

We can make use of an approximation to simplify this expression. If z is small, the logarithmic function can be approximated by:

$$ln(1 + z) = z - z^2/2 + z^3/3 - z^4/4 + \cdots$$

Using the first two terms only, and finding the maximum by differentiating w.r.t. f and setting the result to zero, we have

$$f = (\sum [p_i h(X_i)])/(\sum [p_i (h^2(X_i)]) = r/\sigma^2$$

where r is the mean of the distribution of $h(X_i)$, and σ^2 is close to the variance of $h(X_i)$ – it should be $E[h^2(X_i)]$, which is close to the variance if the mean is small. Thus the Kelly ratio can be applied in the case of stock trading.

9.1.3. Some Properties

We have seen above that maximizing the expected wealth may not be a good idea for an investor. In deriving the Kelly fraction, we are actually maximizing the geometric mean of wealth, or the arithmetic mean of the log of wealth (represented by the gain factor). To summarize the properties of the Kelly criterion, in the long term (i.e. with an infinite sequence), the following propositions would definitely occur (with probability = 1):

- Maximizes the final value of the wealth (compared with any other strategy);
- Maximizes the median of the wealth;
- Minimizes the expected time required to reach a specified goal for the wealth.

However, from the diagrams above we can also see that the path of outcomes can be very volatile. We can state the probability that a certain wealth level is reached $B \text{ x } W > W_0$ before it dips to $A \text{ x } W < W_0$. This is given by (c.f. Sinclair (2013), p.139)

$$P(A, B) = \frac{1 - A^{1-2/f}}{B^{1-2/f} - A^{1-2/f}}$$

For example, if A = 0.5 and B = 2, we get $P(A,B)$ = 2/3, which means that there is 1/3 chance of having the wealth level to become half before it doubles – the risk is very high. Therefore it is common to trade at less than Kelly ratio to reduce the risk.

9.1.4. Advantages and disadvantages

The advantages of adopting the Kelly criterion include the following:

- It is a proportional strategy and will never lose more than the initial wealth.
- On average, this strategy outperforms others.
- Any new trade is a function of the current wealth and is not dependent on previous trades.
- One can control risk by trading at less than Kelly ratio.

Some disadvantages of the method include:

- Maximizing the log utility of wealth may not always be the best option.
- Trade size can be big if the probability of winning is high, hence the trading risk increases.
- The strategy is aggressive and the outcome can be volatile, which is worse for a risk-averse investor.
- A long time is needed before it dominates other strategies.

9.2. Trading strategies with high leverage

9.2.1. Introduction

Many investment strategies are based on employing quantitative methods in the analysis of financial markets which will be used to predict future performance. With ever increasing computational power, a lot of the so-called "quant" strategies are data driven, while the popular strategies can make use of alternative data sets (i.e. data which are not typically used in predicting stock price performance) compared to traditional price data (such as earnings and daily closing prices). Investment banks have tried to sell the product ideas to their institutional customers such as index-linked "Smart Beta" or "Risk Premia" products. Hedge funds are also active in employing these strategies. They tend to make many small bets instead of having concentrated positions. Some examples of these strategies include:

- Fundamental-based – trading decisions are based on some kind of fundamental analysis of the companies involved (but with more quantitative bias); the holding period of the investments can be days or months;
- Convergence or relative value trades based on statistical properties – holding period can be as short as a few minutes up to a few months;
- High frequency trading – very short term trading based on prediction of small movements; holding period of less than one second is common in mature markets such as the United States.

Leveraging strategies are often employed in order to achieve a high return with limited capital. To demonstrate the use of leverage, assume that you have $10000 to buy shares. If share price goes up by 10%, your return is 10%. However, if you can borrow money to invest, the potential return is much higher. For example, you have $10000, but someone lends you an additional $40000, so that you bought $50000 worth of shares. If

share price goes up by 10%, your profit is $5000, and this represents 50% of your original investment.

In the financial markets, there are many ways to achieve the leverage effects. Some of these methods are:

- *Margin accounts (stocks, FX)*. Instead of paying the full price to enter a position, only a certain percentage is required. However, there is a need to maintain a minimum margin level, otherwise there would be a forced termination of the position when losses occur.
- *Use of futures and options*. For example, for HSI futures, we need to pay the equivalent of 1565.4 index points in margin (as of October 3, 2023) to control an asset which is roughly equal to 17000 points.
- *Long/short trades (pairs trading)*. Very high leverage is attainable because it can require very little initial investment. Further details are given in the following sections.

We introduce a term known as *market neutral strategy*. These strategies do not make assumption as to whether the market will rise or fall, and are based on how two assets will move relative to each other. One typical example is called *pairs trading*, which involves entering positions in two assets. For example, we can go long 1000 shares of stock A at $10 per share, and short 2000 shares of stock B at $5 per share on trade date, so that no initial investment is required (other than margins). If share price of A goes up by 10% and share price of B goes up 5%, a total profit of $1000 – $500 = $500 is made. In fact, as long as the relative performance of A is better than B, a profit is earned (e.g. both stocks can go down, but still a profit is possible). The strategy would incur a loss if B outperforms A, irrespective whether there is an upward or downward move in the stock prices.

9.2.2. Pairs trading

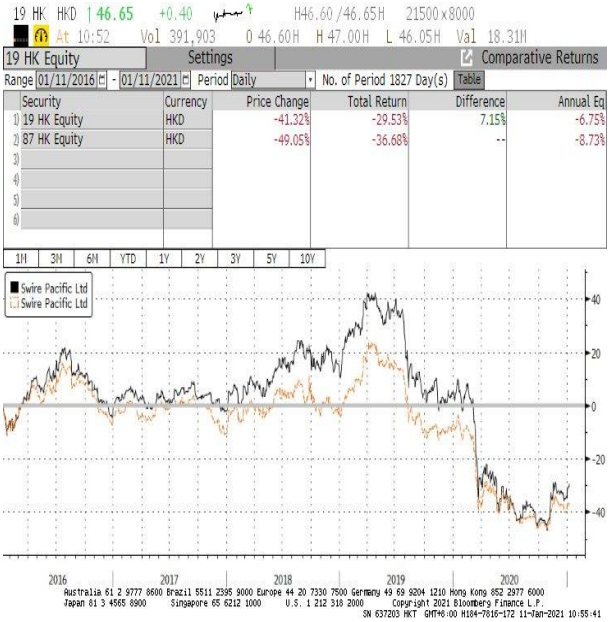
Paris trading is sometimes known as relative-value trading, where the methodology is heavily based on whether two securities will revert back to an underlying long-term relationship between them (a **"mean-reverting"** behaviour). The idea is straightforward: enter a long position in one asset and a short position in another asset. In practice, one needs to consider the cost and possibility of short selling (such as the availability of stock borrowing and regulatory constraints concerning short positions).

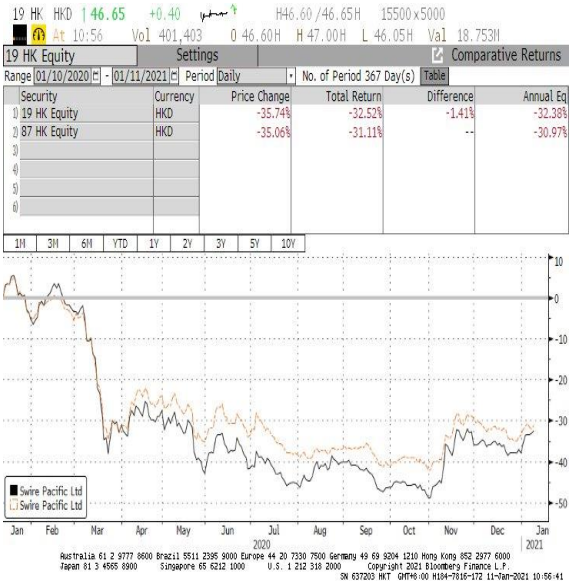
From a trading point of view, the key concern is to establish rules to pick the stock pairs. There are two main groups of strategies. The first type is based on fundamental company characteristics. Some of the typical strategies include:

- Different classes of security for the same company listing in the same market.
- Different listing, such as the local shares against the American Depository Receipt (ADR).
- A company listing simultaneously in different markets, e.g. A-shares vs H-shares.
- Ratio trading based on cross holdings between the companies.
- Outperformance trades based on company characteristics, e.g. industrial sectors.

The second type of strategy will not pay attention to the company's data, but the selection criterion is purely based on past performance. A popular strategy is known as statistical arbitrage, where the time series properties of historical prices would be analyzed. This will be discussed in the next section.

Similar class





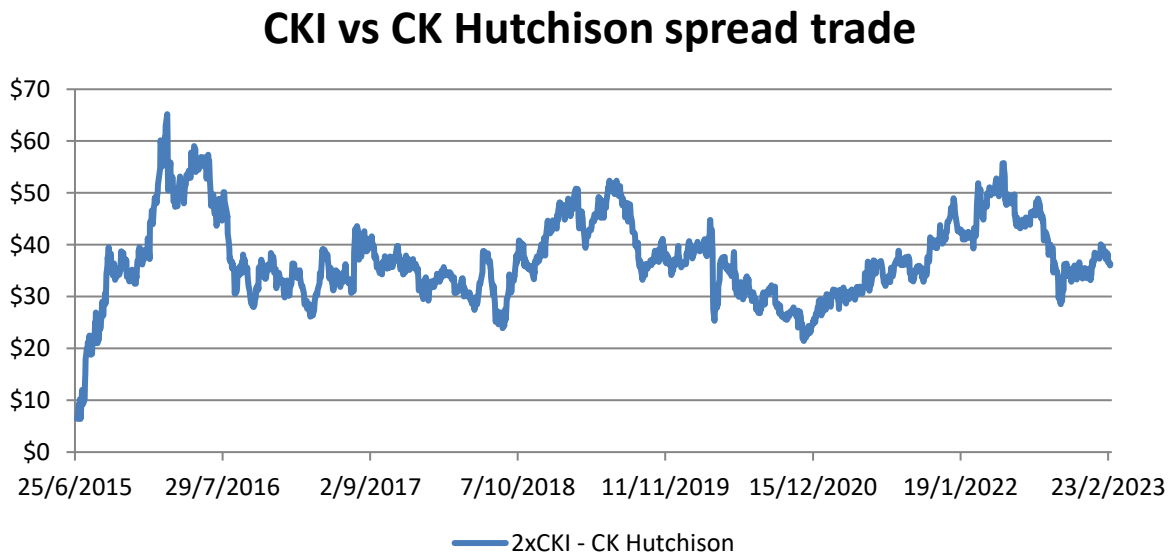
Source: Bloomberg LP

The relative performance of two stocks, Swire Pacific 'A' (0019) and Swire Pacific 'B' (0087), are shown in the diagrams above. Both these stocks are based on the same company, although the shares differ in voting rights and dividend policy. It is noticed that the liquidity of Swire Pacific 'B' is much lower since it is largely owned by the majority shareholder, and the dividend yield of Swire Pacific 'B' is often higher than Swire Pacific 'A'. Theoretically, there should be no difference between the performances of these shares, although short term discrepancy can be seen, mainly due to trading factors. It is thus possible to take advantage of the short term deviation from the fair value and wait for the stock prices to converge. In practice, one side of the trade may not be feasible, as Swire Pacific 'B' is no longer eligible for short selling according to the rules of the Hong Kong Exchange.

Long/short example: premium extraction trade (Source: Bloomberg LP

In the past, Wipro ADR (listed in New York) was usually trading at a premium to the local Indian stock due to foreign ownership rules for the local stock in India. However, the premium level kept changing – within the five year period between 1/2016 and 1/2021, we can see that the premium had a range roughly between 10% and 50% (as seen in the lower graph). A possible trade is as follows: if it is anticipated that the premium will trade higher, one strategy is to (1) borrow the local stock and sell it; (2) convert the Indian Rupee thus received to US Dollar; (3) use the proceeds to buy the ADR. As long as this view of the market is correct, whether the company share price is going up or down is irrelevant. More recently, the premiums of ADRs of Indian stocks seem to have disappeared.

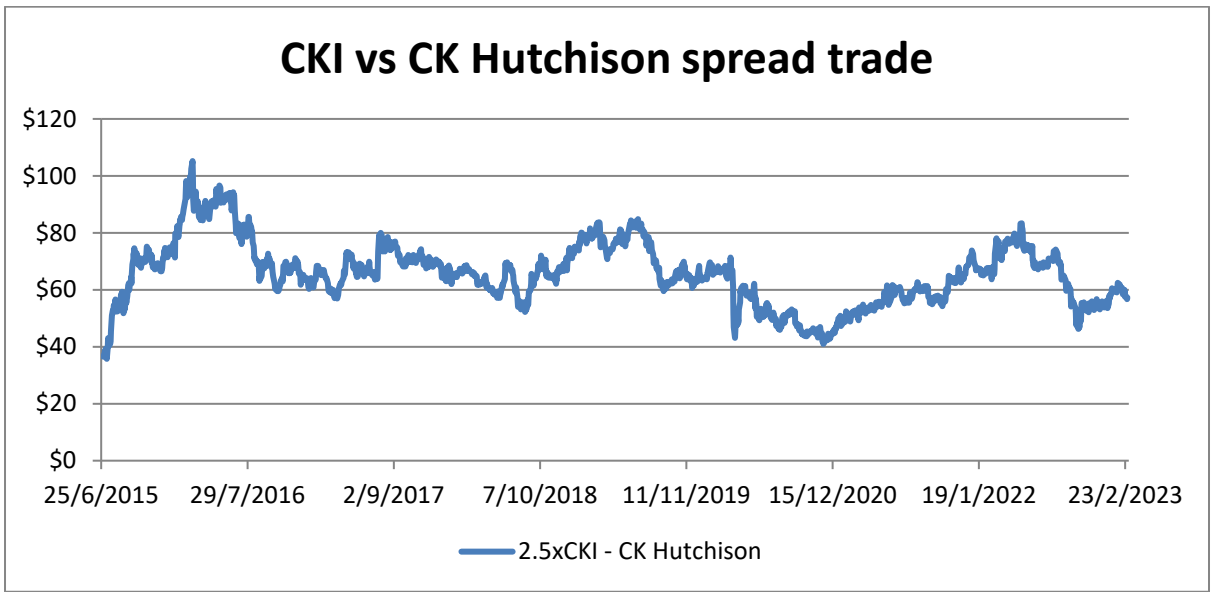
Long/short example: related companies



As of 2022, CK Hutchison (0001) owned 75.67% of Cheung Kong Infrastructure (1038). In March 2023, CK Hutchison's price/earning (P/E) ratio was 5.48 times, with a market capitalization of HKD 182.7 billion. Cheung Kong Infrastructure (CKI) had a P/E ratio of 14.09 times, with a market capitalization of HKD 105.8 billion. While the P/E ratios of companies from different sectors are not comparable, a possible trade idea was to have a

long position in CKI and short position in Hutchison. From the fundamentals given above, a possible ratio was to hold 2 share of CKI and short 1 share of Hutchison.

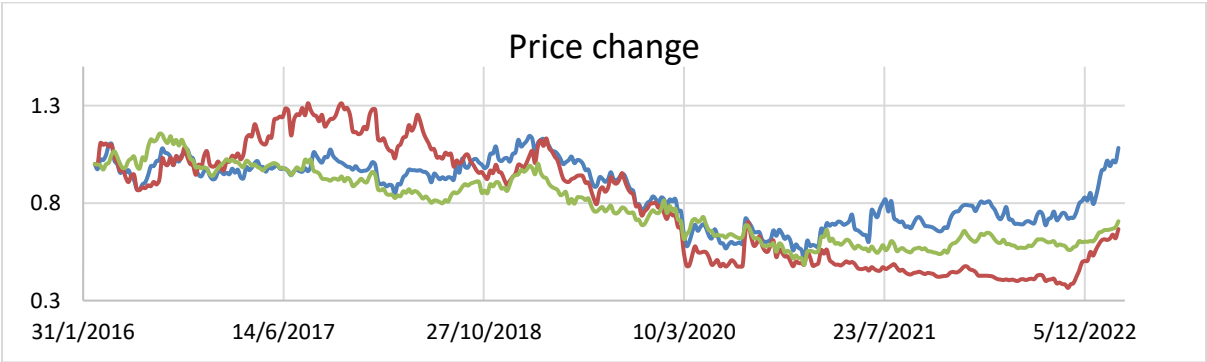
Let's say you put on the trade on June 30, 2015, when the prices of CK Hutchison and CKI were HK$ 113.90 and HK$ 60.20. The cost of trading the strategy = HK$ 2 x 60.2 – 113.9 = $6.50. Other transaction costs should include the borrowing fee for Hutchison, stamp duty, funding cost etc. If transaction costs are ignored and if CKI's price moves up by 1% (0.602) while Hutchison's price remains unchanged, the portfolio return is 2x0.602/6.50 = 18.5%, i.e. the portfolio has a leverage of 18.5x. This trade had been very successful in 2015/6. On March 1, 2016, the spread was worth $65.20. As of March 1, 2023, the spread was $36.30 (Hutchison's price was $47.70, CKI's price was $42.00). Note that this strategy is often used in short term trading instead of for long term investments (dividends have been ignored in the above calculations).

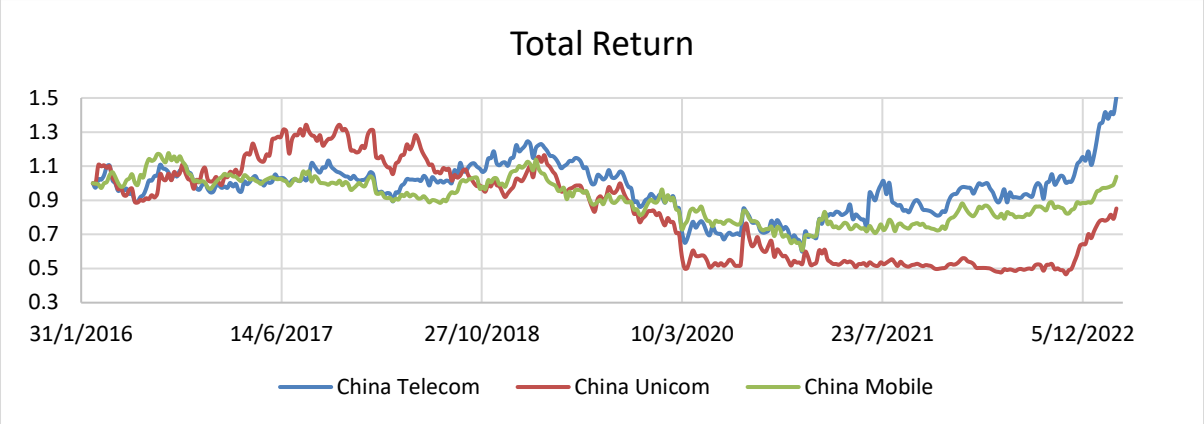


There is no particular reason why a ratio of 2 to 1 must be used. If we use a different ratio, such as long 2.5 shares of CKI and short 1 share of Hutchison, the results are shown in the above diagram. We can see that the profit is less.

Sometimes it is even possible to construct a zero-cost strategy (other than the transaction costs), so that an extremely high leverage could be attained. However, setting the trading ratio is more of an art than a science.

Other pairs trading examples





Another common strategy is to look for outperformance of one company against another within the same sector. For example, we can compare the performance of the telecom companies in China. Three such stocks listing in Hong Kong are China Unicom (0762), China Mobile (0941) and China Telecom (0728). From the diagram above, we can see that China Telecom has outperformed the other two in the last two years (as of March 2023). One can then take a position as to whether this trend will continue. Another possible strategy is to compare the forecasts of companies in the same sector from different countries, such as Citibank vs HSBC. [1]

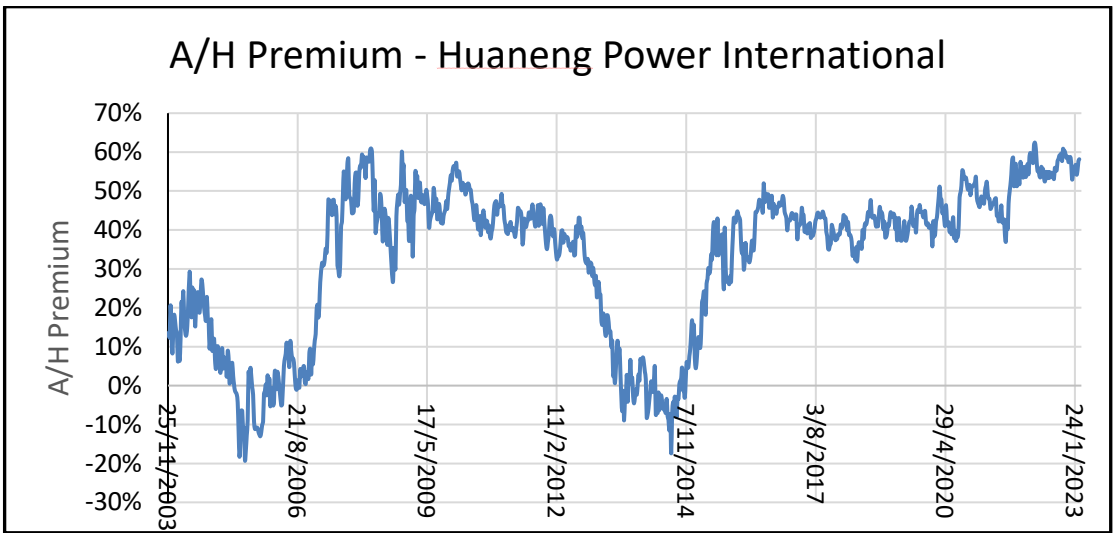
A shares vs H shares

Many Chinese companies are listed both as A-shares in Shanghai / Shenzhen markets and H-shares in the Hong Kong market. However, due to regulatory differences and a different investor base, there has been big discrepancies between the share prices for many companies. For most companies, Chinese share prices have often shown a premium to Hong Kong share prices. Given that the share prices should reflect the fundamentals of the companies, there should not be a consistent big difference in their valuation. On the other hand, there seem to be no effective way of arbitrage, even after the introduction of the Shanghai-Hong Kong Stock Connect (滬港通) in Nov 2014 and the Shenzhen-Hong Kong Stock Connect (深港通) in Dec 2016. The topic has aroused much research interest. A benchmark that can be used to monitor the general trend of the premium between A and H shares is known as the Hang Seng AH Premium Index, as shown in the diagram below (source: https://hk.investing.com, as of March 1, 2023):

[1] See Ganapathy Vidyamurthy, *Pairs Trading: Quantitative Methods and Analysis*, Wiley Finance (2004) for additional details.

An index of 100 means that A-share is trading at par with H-share, whereas a number above 100 indicates that A-share is trading at a premium to the H-share.

Even though A-shares often trade at a premium to the H-shares, the premium level can change over the years. As an example, Huaneng Power International is one of the first batch of Chinese companies listed in Hong Kong as H-shares (listed in January 1998). The graph below shows the premium/discount of it's A-shares over the H-shares between December 2003 to March 2023.



9.2.3. Statistical arbitrage

Another kind of pairs trading strategy does not rely on company fundamentals. Statistical arbitrage uses time series techniques to determine trading signals. Again it is based on the concept of mean reversion, which suggests that the relationship between two securities fluctuates around a constant mean. Whereas the relative behavior between the two securities may deviate from the mean temporarily, it is assumed that the behavior will return to normal after a certain time. This relationship would be quantified, and the typical methods include:

- Measuring the spread or ratio between two securities.
- Regression-based methods.
- Establishing a co-integrated time series.

An arbitrage is supposed to mean that a riskless strategy is conducted by simultaneously trading different assets. On the other hand, there is no guarantee that the relationship must revert back to normal levels. As such, statistical arbitrage cannot be considered as a real "arbitrage"! However, these strategies are very popular in recent years. A detailed discussion is outside the scope of this course.[2] There has been an increasingly large literature based on high frequency trading algorithms and techniques, which can also be considered to belong to this class of strategies.

A simple example

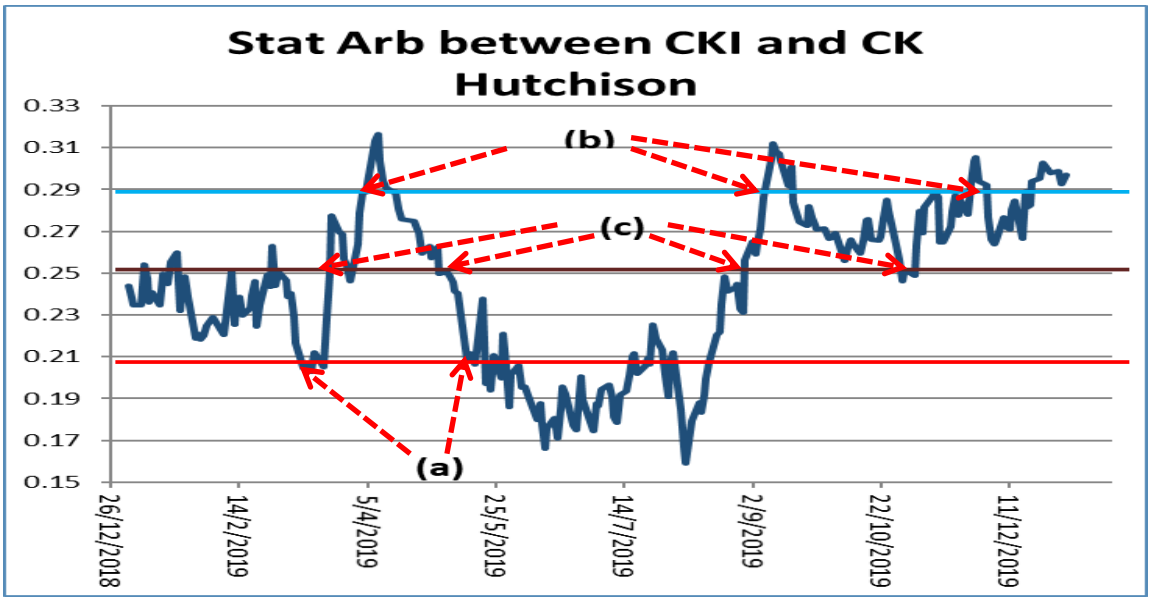
Define the log spread as:

$$s_t = \ln(y_t) - \ln(x_t)$$

where x_t and y_t are the prices of the two stocks at time t. The mean and variance are defined by:

$$\bar{s} = \frac{1}{T}\sum_{t=1}^{T} s_t, \quad \sigma_s^2 = \frac{1}{T-1}\sum_{t=1}^{T}(s_t - \bar{s})^2$$

Simple trading rules can be set up, for example:
If $s_t > \bar{s} + k\sigma_s$, go short in stock y and long in stock x
If $s_t < \bar{s} - k\sigma_s$, go long in stock y and short in stock x
when s_t returns to $\bar{s}$ close out of position



We use historical data for CKI and CK Hutchison to illustrate this trading idea. k has been chosen to be 1.8. (a) represents the signal to sell CKI and buy CK Hutchison; (b) is the signal to buy CKI and sell CK Hutchison; and when (c) is breached the existing position is closed out in order to realize the profit. The trading strategy will have a profit of HKD 12,000 for trading size of 1000 shares each (before transaction cost) over the monitoring

[2] Refer to the appendix at the end of this chapter for an introduction to time series techniques.

period. This is a naïve example which is unrealistic, and real life statistical arbitrage strategies would aim to trade much more frequently while hoping to make a small profit for each trade. Hence it is very important to make use of historical data to train the model in order to identify optimal signals for profitable trades.

Co-integrated time series

A more sophisticated model is to employ the idea of co-integration. Stock prices are often modelled as a random walk process, which is non-stationary. In 1987 Clive Ranger and Robert Engle proposed that for two non-stationary time series x_t and y_t , there may be some kind of relationship between the two: if a specific linear combination becomes a stationary series (say $y_t - \gamma x_t$), the two original series are known as co-integrated. (This idea is crucial in earning the Nobel prize in 2003). More formally, if ε_{xt} and ε_{yt} are white noise processes,

$$y_t = \alpha_y (y_{t-1} - \gamma x_{t-1}) + \varepsilon_{yt}$$
$$x_t = \alpha_x (y_{t-1} - \gamma x_{t-1}) + \varepsilon_{xt}$$

An alternative representation is the *common trends model* from Stock and Watson:

$$y_t = n_{yt} + \varepsilon_{yt}$$
$$x_t = n_{xt} + \varepsilon_{xt}$$

where n_{xt} and n_{yt} are the non-stationary components of the time series and ε_{xt} and ε_{yt} are the stationary components. We can write the co-integrating combination as:

$$y_t - \gamma x_t = (n_{yt} - \gamma n_{xt}) + (\varepsilon_{yt} - \gamma \varepsilon_{xt})$$

Since this combination is stationary, it means that the non-stationary component must be 0, thus:

$$n_{yt} - \gamma n_{xt} = 0 \text{ or } n_{yt} = \gamma n_{xt}$$

Issues in strategy design

The first step in designing an appropriate trading strategy is to identify stock pairs that could potentially be co-integrated. From finance theories such as the APT, stock prices can be related to common factors, so many stocks can potentially satisfy this property. The next step is to conduct a test of co-integration: determine the co-integration coefficient γ and examine the spread series to ensure it is stationary and mean-reverting. Once a satisfactory pair of stocks has been identified, one needs to set up trading rules by defining the entry and exit levels, taking into account of trading costs and anticipated slippage due to market factors.

A successful implementation of the strategy would only be possible based on extensive back-testing of models using historical data.[3] Therefore it is very important to pay attention to data quality. Closing prices may need to be adjusted because of various technical factors such as corporate actions. Furthermore, if a strategy involves comparing stock prices from two different countries, especially for countries at different time zones, there may be a need to make adjustments due to different market hours and holidays.

[3] Back-testing is an important topic and requires careful attention. Problems can come from the small sample size, over-fitting, and survivorship bias. Refer to Lo (2016) for comments on other backtest bias.

Not all companies can engage in pairs trading strategies. In particular, there may be different regulatory concerns. In Hong Kong, short selling is not allowed for every stock.[4] Some types of companies such as mutual funds or retirement funds are governed by ordinances and leveraged trading is not allowed. Banks must fulfill capital adequacy ratios (CARs), with 8% of Tier 1 capital being the global standard, and thus there is a limit to the level of leverage that can be attained. Of course, un-regulated hedge funds are not subject to these restrictions.

9.3. Portfolio insurance

9.3.1. Introduction

Portfolio insurance is a well-known and simple investment strategy which is usually applied to stock portfolios, although it can also be applied to other kinds of risky assets, e.g. mutual funds, or even hedge funds. These are especially popular in the European markets.

Portfolio insurance is a dynamic strategy. It needs to be re-balanced periodically by following some pre-set rules. Basically, when stock goes up, one should allocate more money in stock and less in cash, and vice versa. Therefore it is a kind of "buy high, sell low" approach. The major advantage of these strategies is that they are supposed to provide some kind of protection to the downside risk.

There are two main kinds of strategies: option-based (OBPI) and constant proportion (CPPI). OBPI was popularized since the 1980's, when option trading based on the Black-Scholes model were becoming popular. Many people blamed this strategy as a major cause of the 1987 stock market crash. While OBPI is based on well-known model, this kind of portfolio insurance is less common recently. CPPI works reasonably well under normal circumstances. However, while it is supposed to be protected by a floor, the value could potentially drop below the floor value, if stock prices drop too quickly.

9.3.2. Option-based portfolio insurance (OBPI)

OBPI is based on having a long position of the underlying asset as well as the purchase of a put option. Usually the put option is not purchased outright, but is constructed in a synthetic manner. The idea can easily be illustrated under the Black-Scholes framework. The Black-Scholes equation for European put is:

$$P = -SN(-d_1) + Ke^{-rt}N(-d_2)$$

If we have a long stock plus long put position (assume dividend yield d=0)

$$\begin{aligned} S + P &= S - SN(-d_1) + Ke^{-rt}N(-d_2) \\ &= SN(d_1) + Ke^{-rt}N(-d_2) \end{aligned}$$

This is just a long call position. Therefore, we just need to invest in the following amounts:

$$\begin{aligned} &SN(d_1) && \text{in shares, or } N(d_1) \text{ shares} \\ &Ke^{-rt}N(-d_2) && \text{in cash} \end{aligned}$$

[4] Check the "Designated securities eligible for Short Selling" in http://www.hkex.com.hk under "Market Operations / Securities Trading Information".

stock price	shares	cash
50	0.01	95.78
60	0.04	93.94
70	0.13	88.20
80	0.27	77.29
90	0.45	62.49
100	0.61	46.70
110	0.75	32.53
120	0.84	21.36
130	0.91	13.36
140	0.95	8.04
150	0.97	4.69

- Strike = 100, maturity = 1 year, interest rate = 4%, volatility = 25%.

The table shows the delta trading required when stock prices are at different levels. When share price goes up, we need to hold more shares.

When the stock market crashes (as in 1987), more shares need to be sold in order to maintain the correct distribution of assets, which would push the prices of stocks down further.

9.3.3. Constant Proportion Portfolio Insurance (CPPI)[5]

Features

Fund managers adopting this strategy follow straight rules, i.e. it is non-discretionary. Similar to OBPI, the manager needs to decide the time horizon for trading, and determine a minimum level of return within this period. The strategy will allocate more money in stocks and less in cash when stock price goes up, and vice versa. One known disadvantage is that if there is a heavy and repeated market slump, all the assets would be invested in fixed income instruments until maturity, hence it could not capture any gain even if the market recovers. However, as long as the value of the performing asset does not drop too quickly, the minimum return can be achieved.

Set up

At any point in time before maturity, the CPPI portfolio consists of the following assets:

$$CPPI(t) = RA(t) + L(t)$$

$RA(t)$ is the value invested in the risky asset and $L(t)$ is the value of the protected asset (can be negative if money is borrowed). Initially the total asset level is $CPPI(t) = A(0)$.

The strategy requires the following parameters:

- A minimum asset level, known as the floor value $F(T)$, where T is the intended maturity (must be smaller than the initial asset level $A(0)$).
- a multiplier m, to decide the leverage.
- need to decide what is an appropriate re-balancing frequency, e.g. daily or weekly.

Allocation mechanism

At time 0, $F(0)$ is the present value of the floor. Define the cushion $C(t)$ as: $C(t) = CPPI(t) - F(t)$. Also define the multiplier m via: $RA(t) = m \text{ x } C(t)$. At time $t+1$, the value of the CPPI strategy is:

$$CPPI(t+1) = CPPI(t) + p \text{ x } RA(t) + q \text{ x } L(t)$$

[5] C.f. Black and Perold (1992).

where p= excess % return of the risky asset over one period; q= excess % return of the non-risky asset over one period.

CPPI example[6]

Assume a 5-year CPPI, initial value $A(0)$ = 100, m = 4, interest rate r = 5%, floor = 77.88 (= value of a 5-year zero coupon bond = 100/(1+0.05)^4). At time t=0, cushion $C(0)$ = 100 – 77.88 = 22.12, thus the amount invested in the risky asset = 4 x 22.12 = 88.48, and the amount invested in the non-risky asset is 100 – 88.48 = 11.52.

If the risky asset goes up by 20% in the first month, the value of the strategy is now $CPPI(1)$ = 88.48 x 1.2 + 11.52 x 78.2/77.88 = 117.74 (78.2 is the value of the zero-coupon bond after 1 month = $F(1)$). The new allocation is thus calculated as:

- Cushion $C(1)$ = $CPPI(1) - F(1)$ = 117.74 – 78.2 = 39.54
- Allocation to the risky asset = m x $C(1)$ = 4 x 39.54 = 158.16
- Allocation to the non-risky asset = $CPPI(t) - RA(t)$ = 117.74 – 158.16 = –40.42

Thus this is a leverage position, as money has to be borrowed to invest in the risky asset.

If the risky asset has gone down 20% instead, the calculation will become:

- $CPPI(1)$ = 88.48 x 0.8 + 11.52 x 78.2/77.88 = 82.35
- Cushion $C(1)$ = 82.35 – 78.2 = 4.15
- Allocation to the risky asset = 4 x 4.15 = 16.60
- Allocation to the non-risky asset = 82.35 – 16.60 = 65.75

i.e. only a small proportion is invested in the risky asset when its price drops.

Some additional CPPI features

Additional rules can be imposed on the basic CPPI strategy. These may include:

- Liquidation trigger
  An absolute size limit can be set, such that when the buffer size relative to the total value is too small, the risky position will be liquidated (i.e. $RA(t)$ = 0).
- Leverage cap
  Two kinds of limits are common. For example, there can be a limit on the weight of the risky asset in the total value of the portfolio, or a limit on the size of the total borrowing.
- Variable and 'straight line' floors
  Instead of having a fixed floor level that lasts until the maturity, it is possible to lock in gains through ratcheting floor levels, sometimes with floor levels varying linearly with time.
- Multiples may vary according to the volatility of the asset (sometimes known as dynamic portfolio insurance).

Option on CPPI

If the underlying variable follows a geometric Brownian motion and hedging is done in continuous time, a closed form formula can be found for an option on the CPPI strategy.[7]

[6] The example is taken from Arnaud Gallais, "CPPI Structures on Funds Derivatives", Master Thesis, Royal Institute of Technology, Stockholm (2011).

[7] E.g. M. Escobar, A. Kiechle, L. Seco, R. Zagst, Option on A CPPI, *International Mathematical Forum*, Vol. 6, 2011, no.5, 229-262.

However, it is important to have a clear understanding of the restrictions and the behaviour of the underlying variable. In practice many underlyings are often illiquid, may exhibit jumps, and the re-balancing is performed in discrete time. So the result will be path dependent which makes modelling difficult.

CPPI: advantages and disadvantages

A main advantage of the portfolio insurance strategies is that it is simple to explain and easy to implement, with fixed investment guidelines. Unlike portfolios with option strategies, CPPI requires no upfront cost. Furthermore, the strategy can be implemented on underlyings without liquid option markets, which make it particularly adapted to developing markets or underlying funds.

However, the portfolio return is not determined by the final asset level, but is actually path dependent. Furthermore, while portfolio insurance strategies are supposed to provide protection to the downside risk, it is really not a "true" insurance as *gap risk* still remains. When stock prices move too fast (before re-balancing), it is possible for the portfolio value to drop below the floor.

Assume that re-balancing is carried out within a short period, such that $L(t+1)=L(t)$ (or $q = 0$), $F(t+1)=F(t)$, then $CPPI(t+1) = CPPI(t) + p \times RA(t)$. If $CPPI(t+1)$ does not drop below the floor:

$$CPPI(t) + p \times RA(t) - F(t) > 0$$

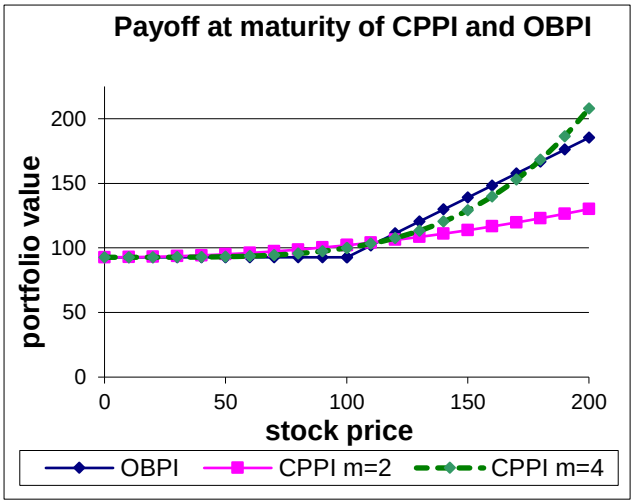
$$\Rightarrow \quad C(t) + p \times m \times C(t) > 0$$

$$\Rightarrow \quad p > -1/m$$

i.e. if the return of the risky asset between re-balancing periods does not drop more than $1/m$ (approximately), the value of the portfolio will not be below the floor. Otherwise, it is still possible to lose more than expected.

9.3.4. Comparing OBPI and CPPI

A_0 =\$100, F_0 =\$80, r=3%, σ=20%, t=1, S_0 = \$100



This diagram is an example of the payoff achievable assuming continuous re-hedging. If re-hedging is infrequent and in the presence of transaction cost, performance is difficult to calculate. In general, higher leverage increases the convexity, but can also mean that there is more probability of underperformance when market drop.

OBPI requires a particular investment horizon (e.g. 1 year), whereas CPPI is supposed to be a continuous trading strategy (until the underlying stock price drops below the floor level), although a time horizon is usually specified. Depending on how the underlying stock price moves, we could not conclude which strategy would always work better. However, when a higher protection level is needed, there is a higher probability that CPPI would outperform OBPI. [8]

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[8] See Betrand and Prigent in Betrand P. & Prigent J-L., "Portfolio insurance strategies: a comparison of standard methods when the volatility of the stock is stochastic", *International Journal of Business*, 8(4), 2003.